

WELBURG SALES MACHINE LEARNING PROJECT

Customer Type Classification, Credited Amount Regression & Customer Segmentation

Project Report

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Dataset	welburg_sales_dataset_2025.xlsx

This report summarizes the executed Jupyter Notebook analysis and presents the dataset, methodology, model results, interpretation, limitations, and future work.

Abstract

This project applies fundamental machine learning techniques to Welburg sales transaction data to investigate customer behavior and sales-related business problems. The notebook contains 5,000 transaction records and 18 original columns covering January to December 2025. Three machine learning tasks were developed: Customer Type classification, Credited Amount regression, and customer segmentation using K-Means. Six classification algorithms, three regression algorithms, and one clustering approach were evaluated. Data quality checks found no missing values and no duplicate rows. Date features were engineered, identifiers and free text were excluded from baseline predictive models, and numerical and categorical preprocessing was implemented through Scikit-learn pipelines. The best classification result by weighted F1 was Decision Tree at 0.4793. Linear Regression was the strongest regression model with an R^2 of 0.6991 and RMSE of 49,940.92. K-Means selected two customer clusters with a silhouette score of 0.3229. The project demonstrates how practical sales data can be structured into measurable machine learning tasks and provides a foundation for future sales analytics and decision-support work.

1. Introduction and Business Problem

Welburg sales transactions contain customer, sales, payment, geographic, and operational information. The project was motivated by practical sales and account-related experience at Welburg Metal, including work with customer and sales-executive information. The analysis asks whether transaction characteristics can support customer classification, credited-amount estimation, and customer segmentation.

Objectives. The project aims to understand sales patterns, prepare data for machine learning, compare classification models for Customer Type, compare regression models for Credited Amount, identify customer segments, evaluate model performance, and interpret the results for future sales analysis.

2. Dataset Description

The executed notebook reports 5,000 records and 18 columns from 1 January 2025 to 31 December 2025. Variables include Date, Transaction ID, Invoice No, Customer Name, Customer Type, Executive, Sales Zone, Sales Channel, Invoice Value, Discount, Sales Amount, Sales VAT, Sales Return, Credited Amount, Payment Method, Bank Name, Remarks, and Month Name.

Customer Type distribution: Retailer 1,386; Wholesale 1,355; Dealer 1,132; Corporate 634; Online 493. The project guideline requires an original/student-created or generated dataset, so the final submission should document the actual collection/generation method truthfully and protect confidential information.

3. Data Preprocessing and Exploratory Analysis

The notebook checks data types, missing values, duplicates, descriptive statistics, and class frequencies. All 5,000 rows are complete, with zero missing values and zero duplicate rows. The Date field was converted to datetime and expanded into Year, Month Number, Quarter, and Day of Week. Identifier fields and Remarks were excluded from the baseline models.

Preprocessing pipeline. Numerical features use median imputation and standardization. Categorical features use most-frequent imputation and one-hot encoding. The pipeline approach ensures that preprocessing is learned from training data rather than the test set.

EDA. The notebook includes numerical distributions, box plots for potential outliers, correlation analysis, customer-type distribution, and customer-type sales summaries. Invoice Value ranges from 5,300 to 499,500, while Credited Amount ranges from 0 to 496,012.38.

4. Supervised Learning — Classification

Customer Type is the classification target. An 80/20 stratified split produced 4,000 training records and 1,000 testing records. Six algorithms were compared.

Model	Accuracy	Precision	Recall	F1
Decision Tree	0.506	0.5193	0.506	0.4793
Random Forest	0.505	0.5292	0.505	0.4741
SVM	0.515	0.4991	0.515	0.4690
KNN	0.408	0.4099	0.408	0.4057
Naive Bayes	0.387	0.4179	0.387	0.3924
Neural Network	0.384	0.3875	0.384	0.3854

Decision Tree achieved the highest weighted F1 (0.4793), while SVM had the highest accuracy (0.515). Five-fold cross-validation of the selected classification pipeline produced a mean weighted F1 of 0.4736.

5. Supervised Learning — Regression

Credited Amount was selected as the regression target. Sales Amount and Sales VAT were excluded to reduce direct target leakage, while identifiers and free text were also removed.

Model	MAE	RMSE	R ²
Linear Regression	34,376.03	49,940.92	0.6991
Random Forest Regressor	36,428.53	54,326.67	0.6439
Decision Tree Regressor	38,323.53	60,273.03	0.5617

Linear Regression was the best regression model on the test set. Its R² of 0.6991 indicates that about 69.9% of the variation in Credited Amount was explained by the predictors used in the notebook. The RMSE of 49,940.92 also shows that monetary prediction error remains substantial.

6. Unsupervised Learning — Customer Segmentation

For clustering, transactions were aggregated at customer level using Total Invoice Value, Average Invoice Value, Total Discount, Total Sales Return, Total Credited Amount, Transaction Count, and Average Credited Amount. The features were standardized before K-Means clustering.

K values from 2 through 8 were evaluated using inertia and silhouette score. The highest silhouette score occurred at K=2. The final silhouette score was 0.3229. This indicates two broad customer groups under the selected feature representation. The clusters should be treated as exploratory segments rather than fixed business labels.

7. Findings and Discussion

The classification task was comparatively difficult, especially for the Online class. The Decision Tree was the strongest model according to weighted F1, although the overall F1 remains moderate. Regression performed more strongly, with Linear Regression achieving an R² close to 0.70. Customer-level aggregation provided a practical basis for segmentation and demonstrated how transaction records can be converted into customer behavior features.

8. Limitations

The analysis covers one year of transaction data and therefore may not represent longer-term seasonal or customer behavior. The customer classes are unequal, and some categories are harder to predict. Available variables do not capture every factor affecting sales and payment behavior. The Remarks field was excluded from baseline modeling despite containing potentially useful business context. The models are academic prototypes and should not be used for operational decisions without further validation. Business and customer confidentiality must also be maintained when sharing the project.

9. Future Work

Future work should expand the historical dataset, add product and quantity information, customer purchase history, payment-delay and return-history features, and apply systematic hyperparameter tuning. Time-aware validation would be appropriate for future sales forecasting. NLP could be applied to transaction remarks after privacy and data-quality review. A Streamlit or Power BI dashboard could present sales trends, customer segments, and validated model predictions to management.

10. Conclusion

This project demonstrates a complete fundamental machine learning workflow using Welburg sales transaction data. Data quality checks, feature engineering, exploratory analysis, pipeline-based preprocessing, supervised classification, supervised regression, and unsupervised clustering were combined in one project. Decision Tree produced the highest weighted F1 for Customer Type classification, Linear Regression produced the strongest Credited Amount regression performance, and K-Means selected two customer clusters. The project therefore demonstrates the practical value of applying machine learning concepts to sales data while also highlighting the need for stronger features, larger historical data, privacy controls, and additional validation before production use.

References

1. National University, Bangladesh. *Fundamentals of Machine Learning — Project Guidelines and Rubric*, 2025.
2. Scikit-learn documentation and API references used for model implementation.
3. Pandas documentation used for data loading and transformation.
4. Matplotlib and Seaborn documentation used for visualization.

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